T3 UV Lagrangian: additional scalar operators

Provenance: saved 16-model operator registry and candidate one-loop RGBeta UV/EFT1 exports. Source beta expressions are algebraically parsed from saved InputForm and typeset in paper coupling notation. No implicit Matchete rescaling. **Status:** candidate tensor basis. General scalar RGBeta–Matchete mapping, quadratic mass-basis matching and full sequential Wilson matching are not certified. *Model-dependent scalar-sector addendum; the original UV, Yukawa and gauge terms are unchanged and documented separately.*

Definition and coverage

The extended BSM scalar sector is $V_{\text{scalar}} = V_{\text{restricted}} + \Delta V_4 + \Delta V_2$. The restricted couplings, including λ_5, are *retained*. ΔV_4 consists of 50 model-dependent complex quartic structures (38 newly named RGBeta vertices plus 12 structurally reused coordinates, counting model occurrences). ΔV_2 catalogues eight possible holomorphic bilinears.

$$\Delta V_4 = \sum_{a\in\mathcal{Q}} \left(\lambda_a \mathcal{I}_a + \lambda_a^* \mathcal{I}_a^\dagger\right), \qquad \Delta V_2 = \sum_{b\in\mathcal{B}} \left(\mu_b^2 \mathcal{J}_b + (\mu_b^2)^* \mathcal{J}_b^\dagger\right).$$

The monomials and invariant tensors below come from the generated registry. They do *not* establish a common Matchete/RGBeta interaction coefficient normalization.

16-model scalar operator inventory

Model	d_{S_1}	d_{S_2}	d_F	α	extra λ	extra m^2
T3 − A ($\alpha = -4$)	1	3	2	-4	0	0
T3 − A ($\alpha = -2$)	1	3	2	-2	5	1
T3 − A ($\alpha = 0$)	1	3	2	0	5	1
T3 − B ($\alpha = -3$)	2	2	1	-3	1	0
T3 − B ($\alpha = -1$)	2	2	1	-1	7	1
T3 − B ($\alpha = 1$)	2	2	1	1	1	0
T3 − C ($\alpha = -3$)	2	2	3	-3	1	0
T3 − C ($\alpha = -1$)	2	2	3	-1	7	1
T3 − C ($\alpha = 1$)	2	2	3	1	1	0
T3 − D ($\alpha = -2$)	3	1	2	-2	5	1
T3 − D ($\alpha = 0$)	3	1	2	0	5	1
T3 − D ($\alpha = 2$)	3	1	2	2	0	0
T3 − E ($\alpha = -4$)	3	3	2	-4	0	0
T3 − E ($\alpha = -2$)	3	3	2	-2	6	1
T3 − E ($\alpha = 0$)	3	3	2	0	6	1
T3 − E ($\alpha = 2$)	3	3	2	2	0	0

Scalar coupling notation

The original topology interaction remains λ_5. Original scalar and portal interactions retain the notation of the earlier analytical reports. Additional quartic structures use λ_7–λ_{12} where these symbols already exist, followed by $\lambda_{13}, \lambda_{14}, \ldots$ for newly named operators. All labels refer to RGBeta potential coefficients in the saved source convention; the Matchete–RGBeta mapping is not fully certified.

Symbol	Scalar field structure	Models
λ_7	$\left(H^\dagger\right)^2 \left(S_1^\dagger\right)^2$	B-1, C-1
λ_8	$\left(H^\dagger\right)^2 S_2^2$	B-1, C-1
λ_9	$\left(S_1^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-1, C-1
λ_{10}	$S_1^\dagger S_2 \left(S_2^\dagger\right)^2$	B-1, C-1

Symbol	Scalar field structure	Models
λ_{11}	$S_1 \left(S_1^\dagger\right)^2 S_2^\dagger$	B-1, C-1
λ_{12}	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	B-1, C-1
λ_{13}	$\left(S_2^\dagger\right)^4$	A-2, D-2, E-2
λ_{14}	$S_2 \left(S_2^\dagger\right)^3$	A-2, D-2, E-2
λ_{15}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	A+0, D+0, E+0
λ_{16}	$\left(S_1^\dagger\right)^2 S_2\, S_2^\dagger$	E+0
λ_{17}	$\left(S_1^\dagger\right)^4$	A+0, D+0, E+0
λ_{18}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2
λ_{19}	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	E-2
λ_{20}	$S_1 \left(S_1^\dagger\right)^3$	A+0, D+0, E+0
λ_{21}	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	B-3, C-3
λ_{22}	$\left(H^\dagger\right)^2 S_1^\dagger\, S_2^\dagger$	A-2, D-2, E-2
λ_{23}	$\left(H^\dagger\right)^2 S_1\, S_2$	A+0, D+0, E+0
λ_{24}	$\left(H^\dagger\right)^2 S_1^2$	B+1, C+1
λ_{25}	$H\, H^\dagger \left(S_2^\dagger\right)^2$	A-2, D-2, E-2
λ_{26}	$H\, H^\dagger\, S_1^\dagger\, S_2^\dagger$	B-1, C-1
λ_{27}	$H\, H^\dagger \left(S_1^\dagger\right)^2$	A+0, D+0, E+0

For distinct SU(2) contractions with the same field content, numbered superscripts/index labels remain independent. These symbols denote model-dependent invariant tensors, not universal Clebsch coefficients.

Additional operators by model

T3 − A ($\alpha = -4$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

T3 − A ($\alpha = -2$)

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2 \left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\, S_1^\dagger \left(S_2^\dagger\right)^2$	1
λ_{22} (new)	$\left(H^\dagger\right)^2 S_1^\dagger\, S_2^\dagger$	1
λ_{25} (new)	$H\, H^\dagger \left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates).

T3 − A ($\alpha = 0$)

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2 S_2 S_2^\dagger$	1
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1 \left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2 S_1 S_2$	1
λ_{27} (new)	$H H^\dagger \left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates).

$\mathrm{T3-B}\,(\alpha = -3)$

Coupling / class	field content	count / basis
λ_{21} (new)	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	1

$\mathrm{T3-B}\,(\alpha = -1)$

Coupling / class	field content	count / basis
λ_{26} (new)	$H H^\dagger S_1^\dagger S_2^\dagger$	2
λ_{10} (reused)	$S_1^\dagger S_2 \left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_9 (reused)	$\left(S_1^\dagger\right)^2 \left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_{11} (reused)	$S_1 \left(S_1^\dagger\right)^2 S_2^\dagger$	1 candidate coordinate
λ_8 (reused)	$\left(H^\dagger\right)^2 S_2^2$	1 candidate coordinate
λ_7 (reused)	$\left(H^\dagger\right)^2 \left(S_1^\dagger\right)^2$	1 candidate coordinate
λ_{12} (reused)	$H H^\dagger S_1^\dagger S_2^\dagger$	1 candidate coordinate

Additional quadratic operators: $S_1^\dagger S_2^\dagger$ (candidate, with Hermitian conjugates).

$\mathrm{T3-B}\,(\alpha = 1)$

Coupling / class	field content	count / basis
λ_{24} (new)	$\left(H^\dagger\right)^2 S_1^2$	1

$\mathrm{T3-C}\,(\alpha = -3)$

Coupling / class	field content	count / basis
λ_{21} (new)	$\left(H^\dagger\right)^2 \left(S_2^\dagger\right)^2$	1

$\mathrm{T3-C}\,(\alpha=-1)$

Coupling / class	field content	count / basis
λ_{26} (new)	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	2
λ_{10} (reused)	$S_1^\dagger\,S_2\,\left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_9 (reused)	$\left(S_1^\dagger\right)^2\,\left(S_2^\dagger\right)^2$	1 candidate coordinate
λ_{11} (reused)	$S_1\,\left(S_1^\dagger\right)^2\,S_2^\dagger$	1 candidate coordinate
λ_8 (reused)	$\left(H^\dagger\right)^2\,S_2^2$	1 candidate coordinate
λ_7 (reused)	$\left(H^\dagger\right)^2\,\left(S_1^\dagger\right)^2$	1 candidate coordinate
λ_{12} (reused)	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	1 candidate coordinate

Additional quadratic operators: $S_1^\dagger\,S_2^\dagger$ (candidate, with Hermitian conjugates).

$\mathrm{T3-C}\,(\alpha=1)$

Coupling / class	field content	count / basis
λ_{24} (new)	$\left(H^\dagger\right)^2\,S_1^2$	1

$\mathrm{T3-D}\,(\alpha=-2)$

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2\,\left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	1
λ_{22} (new)	$\left(H^\dagger\right)^2\,S_1^\dagger\,S_2^\dagger$	1
λ_{25} (new)	$H\,H^\dagger\,\left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates).

$\mathrm{T3-D}\,(\alpha=0)$

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2\,S_2\,S_2^\dagger$	1
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1\,\left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2\,S_1\,S_2$	1
λ_{27} (new)	$H\,H^\dagger\,\left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates).

T3 − D ($\alpha = 2$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

T3 − E ($\alpha = -4$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		

T3 − E ($\alpha = -2$)

Coupling / class	field content	count / basis
λ_{13} (new)	$\left(S_2^\dagger\right)^4$	1
λ_{14} (new)	$S_2\,\left(S_2^\dagger\right)^3$	1
λ_{18} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	2
λ_{19} (new)	$S_1\,S_1^\dagger\,\left(S_2^\dagger\right)^2$	2
λ_{22} (new)	$\left(H^\dagger\right)^2\,S_1^\dagger\,S_2^\dagger$	1
λ_{25} (new)	$H\,H^\dagger\,\left(S_2^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_2^\dagger\right)^2$ (candidate, with Hermitian conjugates).

T3 − E ($\alpha = 0$)

Coupling / class	field content	count / basis
λ_{15} (new)	$\left(S_1^\dagger\right)^2\,S_2\,S_2^\dagger$	2
λ_{16} (new)	$\left(S_1^\dagger\right)^2\,S_2\,S_2^\dagger$	2
λ_{17} (new)	$\left(S_1^\dagger\right)^4$	1
λ_{20} (new)	$S_1\,\left(S_1^\dagger\right)^3$	1
λ_{23} (new)	$\left(H^\dagger\right)^2\,S_1\,S_2$	1
λ_{27} (new)	$H\,H^\dagger\,\left(S_1^\dagger\right)^2$	1

Additional quadratic operators: $\left(S_1^\dagger\right)^2$ (candidate, with Hermitian conjugates).

T3 − E ($\alpha = 2$)

Coupling / class	field content	count / basis
No additional scalar quartic operators in this assignment.		